FarmDtech | Illustrative Business Case

Nigerian agritech research MVP | PYCODE Team | October 2026

Problem and proposed value

Farmers may lack timely access to comparable crop prices and transportation-cost information. FarmDtech makes market comparisons easier by aggregating comparable prices, projecting next-month market trends, and estimating potential net proceeds after user-entered costs.

Illustrative scenario (NOT actual market quotes)

Crop / quantity	Maize / 1,000 kg
Market A assumed price	NGN 500 per kg
Market B assumed price	NGN 570 per kg
Travel assumptions to A	20 km x NGN 200/km + NGN 5,000 fee
Travel assumptions to B	110 km x NGN 200/km + NGN 5,000 fee
Gross at A / B	NGN 500,000 / NGN 570,000
Indicative total costs at A / B	NGN 9,000 / NGN 27,000
Indicative net at A / B	NGN 491,000 / NGN 543,000
Illustrative uplift (B vs A)	NGN 52,000 (approx 10.6%)

What this calculation does and does not show

This is a transparent scenario to demonstrate ROI logic. It is NOT a demonstrated increase in farmers' incomes or actual observed price difference; all prices and distances here are assumptions. Real-life net proceeds depend on verified offers, produce grade, transaction charges, road routing, freight capacity and spoilage.

Validation goals

- Survey farmers and traders to verify the information problem.
- Acquire independently verified weekly market quotes covering selected crops.
- Evaluate forecast error on held-out observed market data; compare with naive baselines.
- Compare recommended versus chosen actual selling-market outcomes.
- Measure farmer usability, alert opt-in and data freshness.

Constraints

The bundled World Bank data are monthly and include modelled estimates, not real-time live quotes. The initial cleaned coverage is 67 market IDs in 15 state labels. Public deployment, database credentials and SMS service integration require team accounts. These constraints must be disclosed in the final submission.